Exp 2

CISCO PACKET TRACER

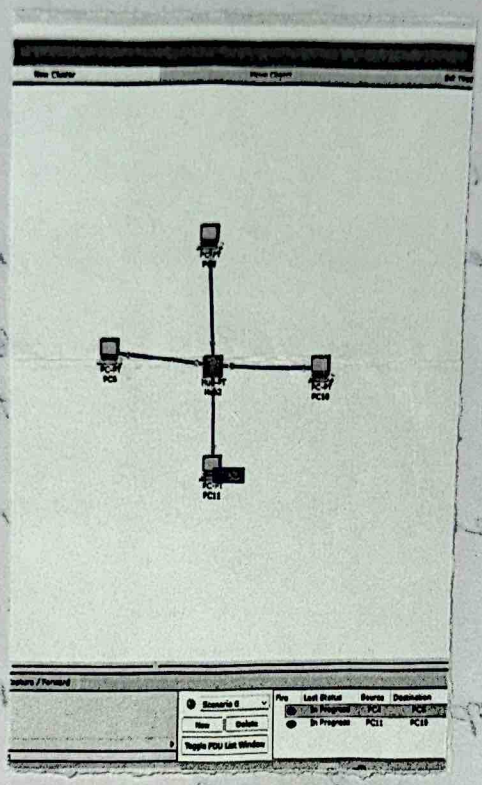
Aim: To study the Packet Tracer tool Installation and User Interface Overview and design a simple computer network using its interface and networking components

STEPS:

* Open the CISCO Packet Tracer
* Observe the UI including the menu bar, Main toolbar, workspace, Common tools Bar, Network component Box, Main toolbar, Workspace, common tools bar
* From the Network component Box, select the required network devices such as PCs, Hub and Switch
* Drag and drop the devices into the workspace.
* Connect the devices using Copper straight-through Cables.
* Configure IP addresses for the PCs using the desktop → IP configuration option.
* Verify that all links are active
* Use the add simple PDU tool to send a packet between two PCs.

Observe packet transmissions in real time mode and simulation mode.

Result: The Cisco Packet Tracer environment was successfully explored and a simple network was designed. The interface components, device connections successfully understood using both Realtime


PC0
PC3
PC10
PC11
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
In Progress
PC11
PC10

Ex 5b1 HUB

Aim : To study the behaviour of a hub by observing how it forwards data packets using cisco packet tracer

Steps :

Open Cisco Packet Tracer

* Drag and place 1 Hub and 4 Generic PCs into the workspace.

* Connect all PCs to the Hub using copper straight - through cables.

* Configure IP addresses for all PCs in the same network.

* Select Add simple PDU from the common Tools bar.

* Send a packet from one PC (source) to another PC (destination)

Result : The Hub broadcasts the received packet to all connected devices. Every PC receives the packet, but the destination accepts it, while the others discard it



23 Exp 8b1 SWITCH

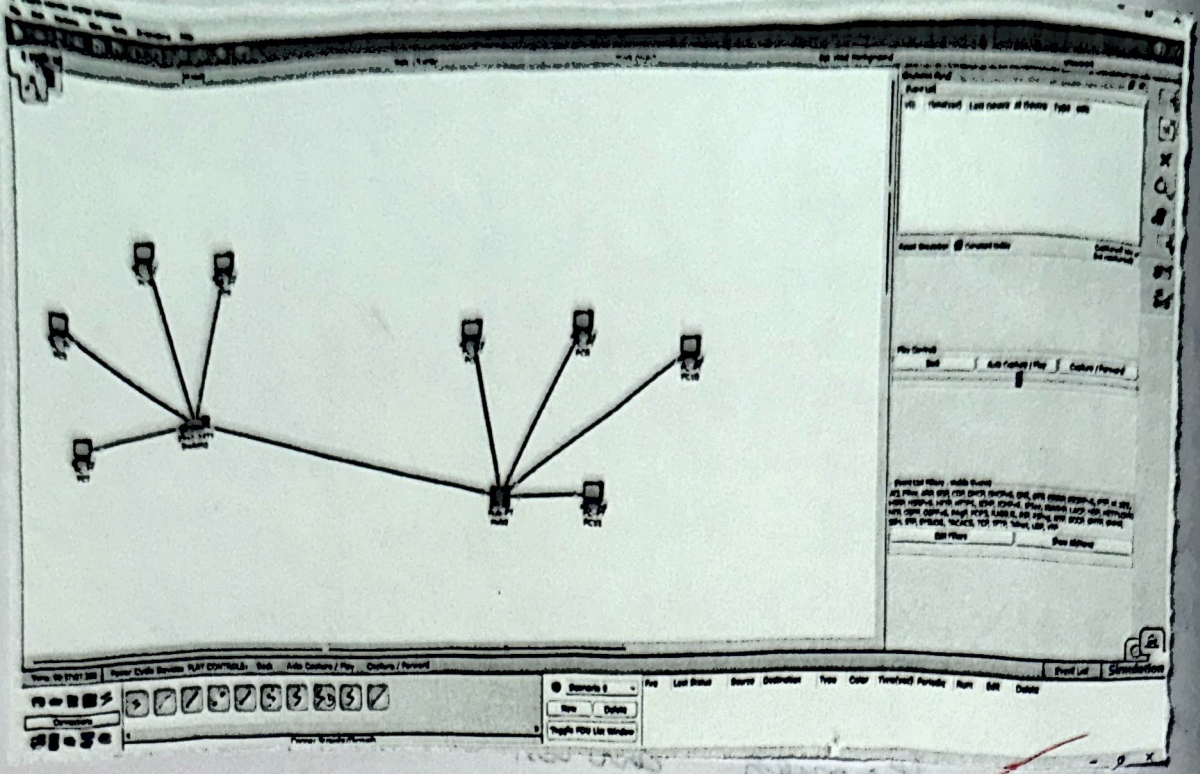
Aim: To study the behaviour of a switch by observing how it forwards data packets using Cisco Packet Tracer.

Steps:

Open Cisco Packet Tracer

* Drag and place 1 switch and 4 generic PCs into the workplace.
* Connects all PCs to the switch using copper straight - through cables.
* Configure IP addresses for all PCs in the same network.
* Select Add Simple PDU from the common tools bar.
* Send a Packet from one PC to another PC

Result: The switch forwards the packet only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and improves communication efficiency.



Exp 3b

HUB AND SWITCH

Aim: To compare the packet forwarding behaviour of a hub and a switch using cisco packet tracer.

Steps:

Create one network using a hub and another using a switch, each connected to 4 PCs.

* configure IP addresses for all PCs in both networks.
* Send a simple PDU between two PCs in each network.
* Observe packet transmission in simulation mode.

Result: The Hub broadcasts packets to all connected devices, whereas the switch forward packets only to the intended destination. Therefore the switch provides better performance, reduced network traffic and improved efficiency compared to the hub.